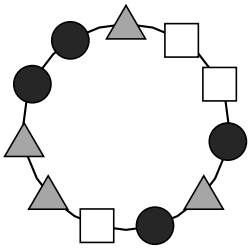
Preview: set-01/by_level/Grade_03-04_Ecolier/y2025_q21 (v4)

Closed loops: groups of neighbours can wrap around the start; recognise the same bracelet turned

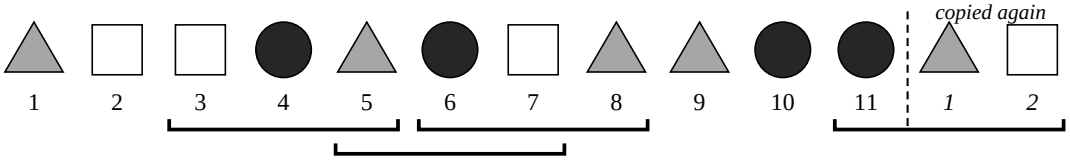
How to think about it: bracelets (stones in a loop)

A bracelet is a **loop**: it has no first and no last stone. When you go around and come back to where you started, the last stone you read and the first one are **neighbours**.



Trick 1: write it in a row, then copy the first stones again at the end

Start at any stone (here: the top one) and go clockwise, drawing each stone in a row. When you are back at the start, draw the first two stones **once more**. Now every group of 3 neighbours, also the one that goes across the start, is just 3 stones side by side in your row.



The brackets show the groups with one circle, one square and one triangle. There are 4 of them, and the last one goes **across the start** (stones 10, 11 and 1). That one is easy to miss!

Trick 2: check each answer picture

- **Count first.** Before: 4 circles, 3 squares, 4 triangles. Lisa takes away one of each, so the answer must have 3 circles, 2 squares and 3 triangles. Pictures with other numbers are out.
- **Counting is not enough!** Some pictures have the right numbers, but the stones taken away were not next to each other. You still have to compare the order.
- **Find a landmark**, a spot that is easy to see, such as two triangles side by side. Find it in both pictures and read clockwise from there. If every stone matches, it is the same bracelet, only turned.

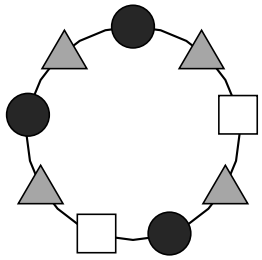
Watch out

- Don't stop at the top: groups can go across the start.
- A turned bracelet is still the same bracelet. Always read clockwise in both pictures.

Try it yourself: lay coins, buttons or Lego bricks in a circle on the table. Take away three neighbours, close the gap and turn the circle a bit. Can you still find where the gap was?

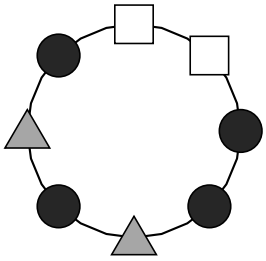
Warm-up: small steps first (no points)

W1. Look at this bracelet with circles, squares and triangles. How many squares are on it?



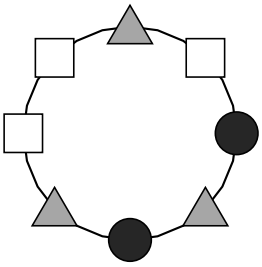
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 7 ***E.*** 9

W2. Going **clockwise** around this bracelet, which three stones come one after another? (Read each answer from left to right.)



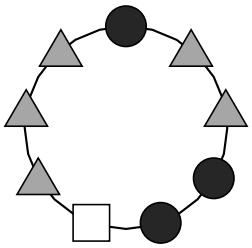
A. □△● ***B.*** ●△□ ***C.*** △□● ***D.*** □●□ ***E.*** ●□□

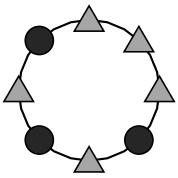
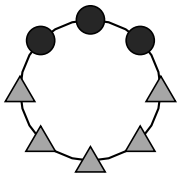
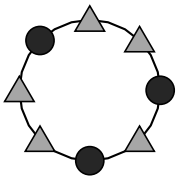
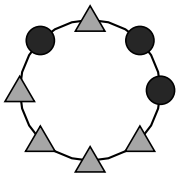
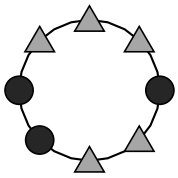
W3. This bracelet has 8 stones. How many different groups of 3 **neighbouring** stones are there? (Don't forget the groups that go across the top.)



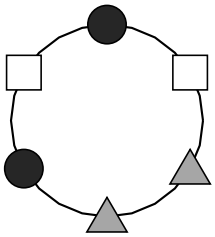
A. 3 ***B.*** 6 ***C.*** 8 ***D.*** 9 ***E.*** 10

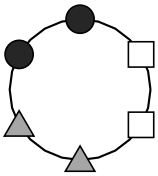
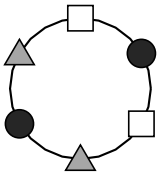
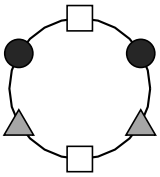
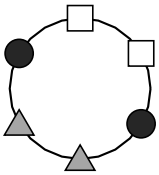
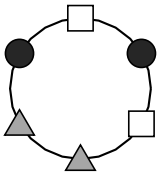
W4. Mia takes the square off this bracelet and closes the gap. Which bracelet is left?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

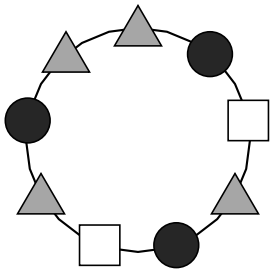
W5. Which picture shows the same bracelet, only turned?

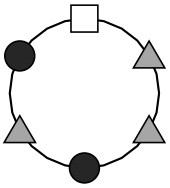
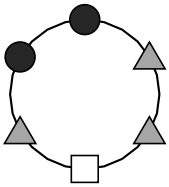
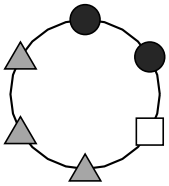
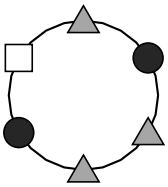
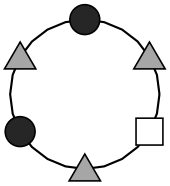


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

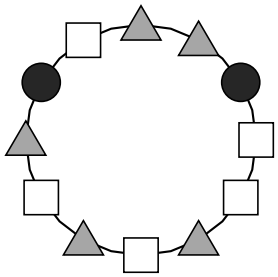
Part A: Each question is worth 3 points

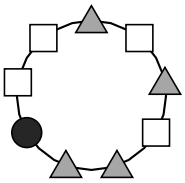
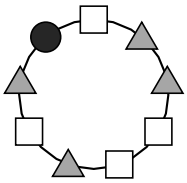
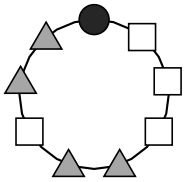
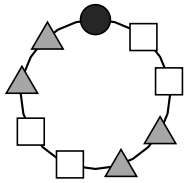
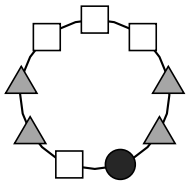
1. The picture shows a bracelet with circles, squares and triangles. Leo removes three neighbouring stones, one of each shape. Which bracelet can Leo get?



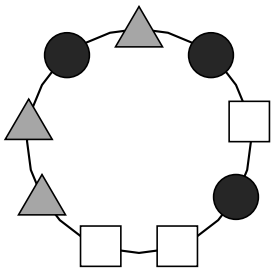
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

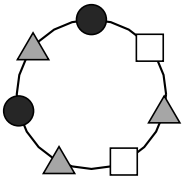
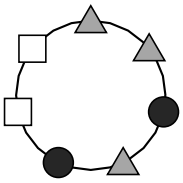
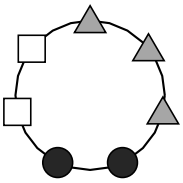
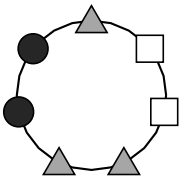
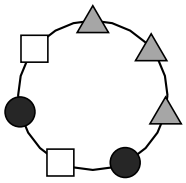
2. The picture shows a bracelet with circles, squares and triangles. Tom removes three neighbouring stones, one of each shape. Which bracelet can Tom get?



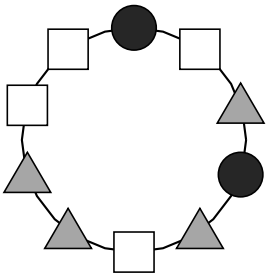
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

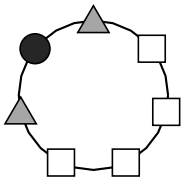
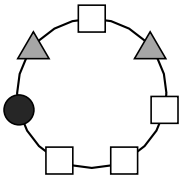
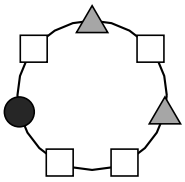
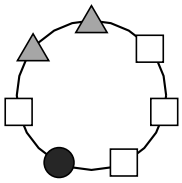
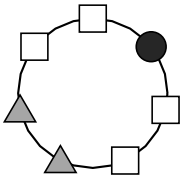
3. The picture shows a bracelet with circles, squares and triangles. Tom removes two neighbouring stones: one square and one circle. Which bracelet can Tom get?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

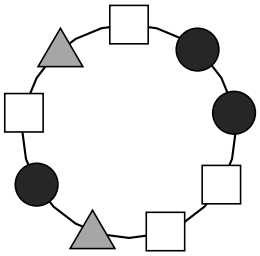
4. The picture shows a bracelet with circles, squares and triangles. Ben removes three neighbouring stones: two triangles and one circle. Which bracelet can Ben get?

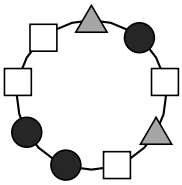
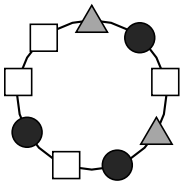
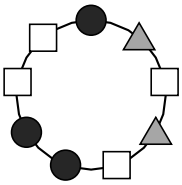
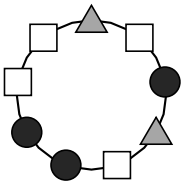
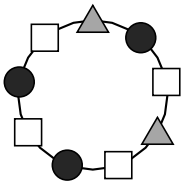


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

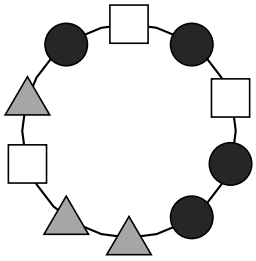
Part B: Each question is worth 4 points

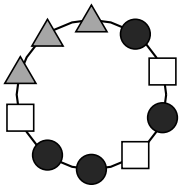
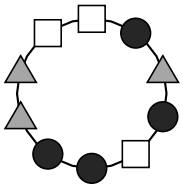
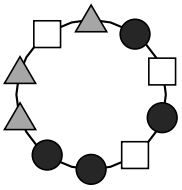
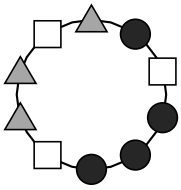
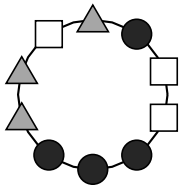
5. Anna has this bracelet with circles, squares and triangles and turns it around a bit. Which picture can show the bracelet now?



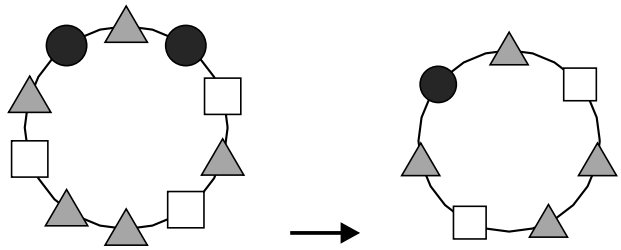
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. Leo has this bracelet with circles, squares and triangles and turns it around a bit. Which picture can show the bracelet now?



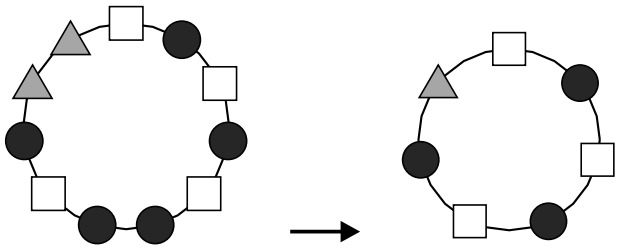
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

7. Anna takes away 3 neighbouring stones from the bracelet on the left. Then the bracelet looks like the one on the right. Which stones did Anna take away?



A. ●△△ ***B.*** ●●△ ***C.*** ●□△ ***D.*** □△△ ***E.*** ●●□

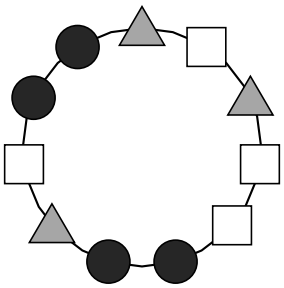
8. Ben takes away 4 neighbouring stones from the bracelet on the left. Then the bracelet looks like the one on the right. Which stones did Ben take away?



A. ●●●□ ***B.*** ●●□□ ***C.*** ●□△△ ***D.*** ●●□△ ***E.*** ●●●△

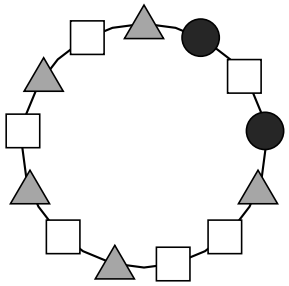
Part C: Each question is worth 5 points

9. Ben has this bracelet with circles, squares and triangles and wants to take away three **neighbouring** stones, one of each shape. In how many different places of the bracelet is this possible?



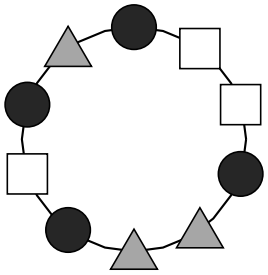
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

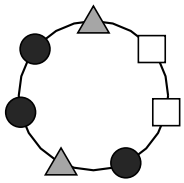
10. Tom has this bracelet with circles, squares and triangles and wants to take away three **neighbouring** stones, one of each shape. In how many different places of the bracelet is this possible?

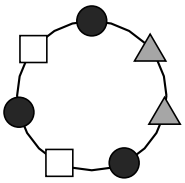


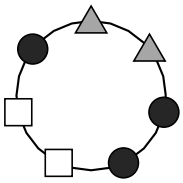
A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

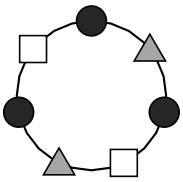
11. The picture shows a bracelet with circles, squares and triangles. Leo removes three neighbouring stones, one of each shape. Which bracelet can Leo **not** get?

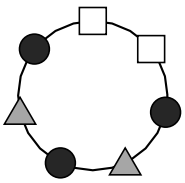


A. 

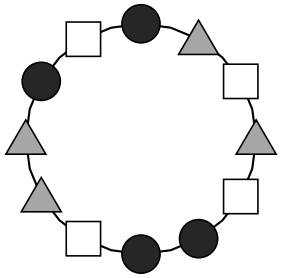
B. 

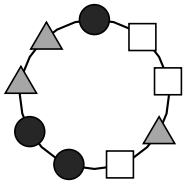
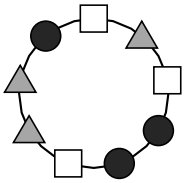
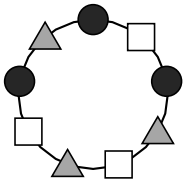
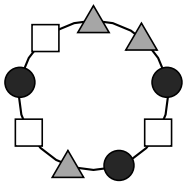
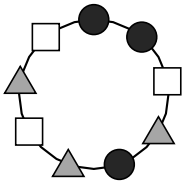
C. 

D. 

E. 

12. The picture shows a bracelet with circles, squares and triangles. Sara removes three neighbouring stones, one of each shape. Which bracelet can Sara **not** get?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Answer Key

W1.	**B**	**W2.**	**E**	**W3.**	**C**	**W4.**	**E**	**W5.**	**E**	**1.**	**D**
2.	**A**	**3.**	**B**	**4.**	**E**	**5.**	**A**	**6.**	**C**	**7.**	**C**
8.	**D**	**9.**	**C**	**10.**	**C**	**11.**	**A**	**12.**	**A**		

Solutions

W1. B (2) — Put your finger on the top stone and go around once, counting the squares: 2.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · w_count]

W2. E — Go around clockwise and look at every group of 3 neighbours. The right group goes across the top: the last stone(s) before the top and the first one(s) after it. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · w_row_of_three]

W3. C (8) — Every stone starts one group: that stone and the next two, going clockwise. 8 stones, so 8 groups. In a straight row there would only be 6. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · w_groups_count]

W4. E — The other stones stay in the same order. Find a landmark (for example two equal stones side by side) and read clockwise: only E matches, turned a bit. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · w_remove_one]

W5. E — Find a landmark in the bracelet (for example two equal stones side by side) and read clockwise from there. Only E matches all 6 stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · w_turned]

1. D — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group circle, square, triangle leaves the bracelet in D, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_three]

2. A — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group square, triangle, circle leaves the bracelet in A, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_three]

3. B — Go around the bracelet and look at every group of 2 neighbours (also the ones across the top). Taking away the group circle, square leaves the bracelet in B, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

4. E — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group triangle, circle, triangle leaves the bracelet in E, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

5. A — Pick a special spot (for example a pair of equal stones) and read the stones in order from there, going around. Only A has all 9 stones in the same order; in the others two stones have swapped places.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · same_bracelet]

6. C — Pick a special spot (for example a pair of equal stones) and read the stones in order from there, going around. Only C has all 10 stones in the same order; in the others two stones have swapped places.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · same_bracelet]

7. C — Count each shape before and after: circles 2 → 1, squares 3 → 2, triangles 5 → 4. So Anna took away 1 circle, 1 square and 1 triangle. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · which_removed]

8. D — Count each shape before and after: circles 5 → 3, squares 4 → 3, triangles 2 → 1. So Ben took away 2 circles, 1 square and 1 triangle. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · which_removed]

9. C (3) — Go once around the bracelet and look at each group of 3 neighbours. Don't forget the group across the top: 1 of them. In all 3. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · count_places]

10. C (4) — Go once around the bracelet and look at each group of 3 neighbours. Don't forget the group across the top: 1 of them. In all 4. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · count_places]

11. A — Find all groups of 3 neighbours with one stone of each shape (also across the top) and take each one away. A has the right stones, but they were not next to each other. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · cannot_make]

12. A — Find all groups of 3 neighbours with one stone of each shape (also across the top) and take each one away. A has the right stones, but they were not next to each other. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · cannot_make]